English Medium

EDU TERIA

72nd BPSC

Prelims Online Batch

BIOLOGY

Topic wise Notes

LECTURE 32

Molecular Basis of Inheritance, Branches of Biology

Molecular Basis of Inheritance, Brahcnes of Biology

GENETIC CODE

- Term Given by George Gamow.
- Discovered by Nirenberg, Matthaei and Khorana.
- The relationship between the sequence of amino acids in a polypeptide chain and nucleotide sequence of DNA or m-RNA is called genetic code.
- There occur 20 types of amino acids which participate in protein synthesis.
- DNA contains information for the synthesis of any types of polypeptide chain.
- In the process of transcription, information is transferred from DNA to m-RNA in the form of complementary N2-base sequences.
- m-RNA contains code for each amino acid and it is called codon.
- A codon is the nucleotide sequence on-m-RNA which codes for particular amino acid; whereas the genetic code is the sequence of nucleotides on m-RNA molecules, which contains information for the synthesis of polypeptide chain.

Triplet Code: -

- The main problem of genetic code was to determine the exact number of nucleotide in a codon which codes for one amino acid.
- There are four types of N2 -bases in m–RNA (A, U, G, C) for 20 types of amino acids.
- If genetic code is singlet i.e. codon is the combination of only one nitrogen base, then only four codons are possible A, C, G and U.
- These are insufficient to code for 20 types of amino acids.
- If genetic code is doublet (i.e. codon is the combination of two nitrogen bases) then 16 codons are formed.
- 16 codons are insufficient for 20 amino acid.
- Gamow (1954) pointed out the possibility of three letters code (Triplet code).
- Genetic code is triplet i.e. one codon consists of three nitrogen bases
- In this case there occurs 64 codons in dictionary of genetic code.
- 64 codons are sufficient to code 20 types of amino acids.

First Base ↓/ Second Base →	U	C	A	G
U	UUU - Phe UUC - Phe UUA - Leu UUG – Leu	UCU - Ser UCC – Ser UCA - Ser UCG - Ser	UAU – Tyr UAC – Tyr UAA - Stop UAG - Stop	UGU - Cys UGC – Cys UGA - Stop UGG - Trp
C	CUU-Leu CUC – Leu CUA - Leu CUG - Leu	CCU - Pro CCC- Pro CCA - Pro CCG - Pro	CAU - His CAC – His CAA - Gln CAG – Gln	CGU - Arg CGC – Arg CGA - Arg CGG – Arg
A	AUU - Ile AUC – Ile AUA - Ile AUG – Met (Start)	ACU - Thr ACC – Thr ACA - Thr ACG - Thr	AAU - Asn AAC – Asn AAA - Lys AAG – Lys	AGU - Ser AGC– Ser AGA - Arg AGG – Arg
G	GUU-Val GUC – Val GUA - Val GUG – Val	GCU - Ala GCC – Ala GCA - Ala GCG - Ala	GAU – Asp GAC – Asp GAA - Glu GAG – Glu	GGU - Gly GGC – Gly GGA - Gly GGG – Gly

Codons(64)		
3 codons ↓ Stop codons	Initiator codon/Starting condon	60 codons for all 20 types of amino acid
UAG UGA UAA	(AUG) ↓ Methionine can also indicate	Starting codon

Genetic Laws:

The fundamental laws of genetics were proposed by Gregor Johann Mendel based on his experiments on pea plants. These laws explain the inheritance of characters from parents to offspring.

Three laws:

1. Law of Dominance
2. Law of Segregation
3. Law of independent assertment

Monohybrid	Dihyboid	
(AABbCC)	(AaBbCC)	
Trihybrid	Height	
AaBbCc	**Dwarf**	**Tall**
	TT geno type	TT Tt genotype

1. Law of Dominance

When two contrasting alleles are present together in an organism, only one expresses itself in the F_1 generation, while the other remains hidden.

Expressed factor → Dominant trait

Hidden factor → Recessive trait

Monohybrid Cross	
Duro	Dwarf Plant
Tall Plant TT	TT

2. Law of Segregation

Height – TT Tt tt

They get segregated during go met formation During gamete formation, the two alleles of a character separate from each other so that each gamete receives only one allele. It is also called the: Law of Purity of Gametes

Dihybrid Cross	
Tall and Purple	Dwarf and white
TTPP	ttpp

3. Law of Independent assortment

The alleles of different genes assort independently during gamete formation and are inherited independently of one another.

Example: Seed shape and seed color in pea plants:

- Round (R) / Wrinkled (r)
- Yellow (Y) / Green (y)
- Cross: RRY Y × rryy

F_2 generation shows new combinations:

- Round Yellow
- Round Green
- Wrinkled Yellow
- Wrinkled Green
- Phenotypic ratio: 9 : 3 : 3 : 1

EUBACTERIA

- They were first observed in rainy water and later in teeth scum by Leeuwenhoek (1675) and called them "Animalcule".
- F.J. Cohn and Ehrenberg first of all coined the name "Bacteria".
- Bergey placed bacteria in "Prosophyta group" and wrote a book "Manual of Determinative Bacteriology".
- This book is known as "Bible of bacterial classification".

NUTRITION IN BACTERIA

- Compared to many other organisms, bacteria as a group show the most extensive metabolic diversity.
- Most of the bacteria are heterotrophic but some are autotrophic. On the basis of nutrition bacteria are classified into following three categories.
- **Autotrophs:** These bacteria use light or chemical energy for their own food synthesis.

On the basis of source of energy autotrophs are of following two type :

Photosynthetic autotrophs

- These bacteria use light energy for food synthesis.
- Photosynthetic pigments are present in cytoplasm for photosynthesis.
- In these bacteria photosynthesis is non oxygenic.
- Some photosynthetic bacteria are: -
  - Purple Sulphur bacteria - e.g. Chromatium
  - Green Sulphur bacteria - e.g. Chlorobium, Thiothrix
  - Purple non-Sulphur bacteria – e.g. Rhodospirillum, Rhod pseudomonas

Chemosynthetic autotrophs –

- They use chemical energy instead of light energy for food synthesis.
- Chemical energy is obtained by oxidation of chemical compounds (inorganic or organic).
- e.g. Nitrifying bacteria - They oxidize nitrogenous compounds and obtain energy.

Heterotrophs

- Most of the bacteria are heterotrophic, i.e., they cannot manufacture their own food.
- The majority of heterotrophic bacteria are important decomposers.
- They are useful in making curd from milk, production of antibiotics, fixing nitrogen in legumes.
- Some are pathogens to human beings, animals and plants.

- They receive their own food from dead organic matter or living organisms.
- These are of following types—
  - **Saprotrophic bacteria** – These bacteria obtain food from dead and decaying organic matter. e.g. Bacillus vulgaris, Clostridium botulinum, Pseudomonas, Staphylococcus
  - **Parasitic bacteria** – They obtain their food from living organism e.g. Mycobacterium leprae, Mycobacterium tuberculosis

BACTERIA

USEFUL ACTIVITIES:

- **Ammonification - Ammonifying bacteria**
  - Some bacteria convert Protein (present in decaying plants & animals) into Ammonia. e.g., Bacillus vulgaris
- **Nitrification Nitrifying bacteria –**
  - These bacteria convert Ammonia into Nitrite and later into Nitrate. NH3 Nitrosomonas NO_2 (Nitrite) Nitrobacter NO_3 (Nitrate)

USEFUL ACTIVITIES:

- **Nitrogen fixation-Nitrogen fixing bacteria**
- These bacteria convert the atmospheric nitrogen into ammonia and then into nitrogenous compounds like amino acids, nitrate or ammonium salts.
- Nitrogen fixation is done by two methods –

A. **Symbiotically** - Some bacteria live symbiotically and exhibit nitrogen fixation.
  - **Rhizobium -** found in the root nodules of legumes
  - **Azorhizobium -** found in the stem nodules of Sesbania plant

A. Symbiotically - Some bacteria live symbiotically and exhibit nitrogen fixation.
  - **Azospirillum -** Found on root surface of cereals (e.g. Wheat, Rice, Maize) i.e., superficial symbiosis.
  - **Frankia (Filamentous bacteria or actinomycetes) -** It is found in root nodules of non leguminous plant Casuarina and Alnus plants.

B. A symbiotically
  - Some bacteria are found freely in soil and perform nitrogen fixation. e.g. Clostridium, Chromatium, Azotobacter, Azospirillum, Beijernickia Rhodomicrobium, Rhodospirillum, Rhodopseudomonas

Note: Azotobacter and Beijernickia are aerobic Rhodospirillum is anaerobic bacteria.

- Both Rhizobium and Frankia are free living in soil, but as symbionts, can fix atmospheric nitrogen.

Antibiotics

- For example, streptomycin is obtained from Streptomyces griseus (It is an actinomycetes)
- Term antibiotic was given by S.A. Waksman
- First discovered antibiotic from bacteria was streptomycin.
- Many antibiotic medicines are obtained from bacteria.
- Some substances produced by microorganisms which inhibit the growth of

other micro-organisms are called antibiotic substances.

Many bacteria are used in industries :

- Vinegar formation (acetic acid)–
- Ethanol Acetobacter acetic acid
- Retting of fibers – Separation of plant fibers by the help of bacteria e.g. Clostridium, Butyric acid bacteria
- Flavouring / curing of tea leaves and processing of tobacco leaves - e.g. Bacillus megatherium, Micrococcus consistence

Production of Vitamins

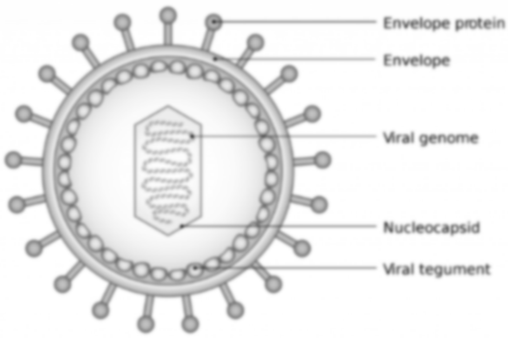
- Clostridium butylicum produces → Riboflavin (Vit. B2) and Butyric acid
- Propionibacterium and Bacillus megatherium produce - Vit. B12
- coli (coliform bacteria) produces → Vit. E., Vit. K.
- coli bacteria found in alimentary canal of human beings.

BACTERIA

- **Purity of Ganga water** - In Gangetic water, a bacteria Bdellovibrio bacteriovorus is found, they kill the other water polluting bacteria.
- **Pollution indicates bacteria-** Water in which E. coli bacteria are present known as polluted water.
- Quality of water depends on number of E. coli.
- If E. coli is very much in no. the water will be highly polluted.
- So, E. coli is known for pollution indicating bacteria.
- **Bacteria for genetic engineering -** Eg. E. coli and Agrobacterium → These are Gram (-) bacteria.

VIRUS

- Viruses are non-cellular, microscopic, and infectious agents that exist as inert crystalline structures outside living cells.
- They are obligate parasites that act as nucleoproteins-containing genetic material (RNA or DNA) enclosed in a protein coat (capsid)—and cannot reproduce outside a host.



- **Non-cellular Entity:** They lack cellular structure, so they are not included in the five-kingdom classification.
- **Structure:** They consist of a capsid (protein shell) made of subunits called capsomeres, surrounding genetic material.
- **Genetic Material:** Viruses contain either RNA or DNA, never both.
- **Host Specificity:** They infect plants, animals, and microorganisms (bacteriophages).
- **Nature:** They are active only inside a living cell, using the host's machinery to replicate and often killing the host

Based on Host:

- **Plant viruses:** Usually contain single-stranded RNA. Example: Tobacco Mosaic Virus (TMV).
- **Animal viruses:** Contain single or double-stranded RNA or double-stranded DNA.
- **Bacteriophages:** Viruses that infect bacteria; typically have double-stranded DNA.
- **Structure:** Viruses can be rod-shaped (TMV), polyhedral/cuboidal (Adenovirus), or tadpole-shaped (Bacteriophage).
- **D.J. Ivanowsky (1892):** Recognized certain microbes as the causal agent of mosaic disease in tobacco. M.W.
- **Beijerinck (1898):** Demonstrated that the extract of infected tobacco plants could cause infection in healthy plants and named the fluid Contagium vivum fluidum (infectious living fluid).
- **W.M. Stanley (1935):** Showed that viruses could be crystallized, and crystals consist largely of proteins.
- **Viroid's:** Smaller than viruses, they lack a protein coat and have free RNA (discovered by T.O. Diener).

- **Prisons:** Abnormally folded proteins that cause infectious neurological diseases.

Branches of Biology

Biology is a broad field, and it is divided into several branches based on what aspect of life is being studied. Here are the main branches of biology:

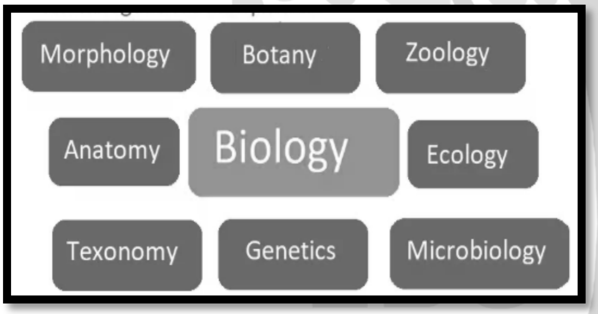
Major Branches of Biology

I. **Botany :** Focuses on the study of plants, including their structure, growth, reproduction, and classification.

II. **Zoology :** Deals with animals—their behavior, physiology, classification, and distribution.

III. **Microbiology :** Studies microscopic organisms such as bacteria, viruses, fungi, and protozoa.

IV. **Genetics :** Examines how traits are passed from parents to offspring and how genes function.



V. **Ecology :** Looks at how living organisms interact with each other and their surroundings.

VI. **Anatomy :** Focuses on the physical structure of organisms and their parts.

VII. **Physiology :** Deals with how the body and its organs function.

VIII. **Cell Biology :** Studies the structure and function of cells, the basic unit of life.

IX. **Molecular Biology :** Focuses on DNA, RNA, proteins, and their roles in life processes.

X. **Evolutionary Biology :** Examines how species change over time through evolution.

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